# Forechecking Pressure Topology
## A Continuous Geometry Framework for Retrieval Denial

Big Data Cup 2026 Final Manuscript

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**Abstract**

Forechecking systems are typically communicated with discrete tactical labels, but retrieval denial is produced by continuous spatial interactions over time. We introduce **Retrieval Denial Topology (RDT)**, a carrier-centric framework that quantifies pressure intensity, corridor availability, first-3-second lane closure, and role-layer coordination (F1/F2/F3 with D-pinch support). Using all 10 Big Data Cup 2026 public games at 5v5, we segment 2,086 defensive-zone retrieval episodes and evaluate 14,921 tracking frames. The framework shows a monotonic pressure-response relationship for denial (46.1% in the lowest pressure decile to 64.1% in the highest), and materially outperforms nearest-defender/count baselines for D-zone turnover prediction (best AUC 0.742 vs 0.640 nearest-defender baseline). We further provide coach-native archetypes, score-state context splits, and team process benchmarking that support practical pre-scout and clip-review deployment.

**Keywords:** forechecking, puck retrievals, spatial topology, lane closure, hockey analytics

# 1 Introduction

Forechecking decisions are often discussed as discrete systems (for example, 1-2-2 or 2-1-2), while execution quality is continuous and timing-dependent. On defensive-zone retrievals, the coaching problem is not only "which system was called," but how quickly viable exit lanes are removed and whether pressure layers are synchronized.

This paper uses a single organizing hypothesis: **retrieval denial is lane-removal dynamics**. We model this dynamic with a carrier-centric pressure topology and evaluate whether it is simultaneously interpretable for coaching and predictive for high-value defensive outcomes.

# 2 Data and Episode Construction

## 2.1 Dataset

The analysis uses all 10 Big Data Cup 2026 public games, combining synchronized tracking, event logs, and orientation metadata. The modeling unit is a 5v5 defensive-zone retrieval episode.

## 2.2 Episode Segmentation and Outcome Definitions

Episodes begin when possession is secured in the defensive zone under 5v5 and end at the first terminal condition: controlled exit allowed, forced dump-out, turnover in defensive zone, stoppage, or timeout guardrail. Segmentation is rule-based and fully auditable.

Table 1: Dataset Scope and Outcome Base Rates

| Metric | Value |
|---|---|
| Episodes | 2,086 |
| Tracking frames sampled (stride = 20) | 14,921 |
| Episodes with complete role-aware feature set | 1,569 |
| Controlled exits allowed | 39.9% |
| Forced dump-outs | 28.6% |
| Turnovers forced in D-zone | 26.1% |
| Stoppage + timeout | 5.5% |

# 3 Methods

## 3.1 Retrieval Denial Topology (RDT)

At each sampled frame, the likely puck carrier is inferred as the nearest offensive skater to puck location. Defender pressure at point $x$ is:

$$P(x) = \sum_i \exp\left(-\frac{\|x - x_i\|^2}{2\sigma_i^2}\right) \exp\left(-\frac{\mathrm{ETA}_i(x)}{\tau}\right).$$

The implemented pass uses acceleration-constrained reachability (bounded-acceleration ETA approximation) and anisotropic directionality (heading-aware spread and turning penalties).

## 3.2 Core Definitions

**Denial** is defined as turnover in defensive zone or forced dump-out. **Corridors** are middle lane, strong boards, weak boards, and behind net. The **cheapest corridor** is the minimum-cost instantaneous exit option. **3-second closure rate** is the drop in cheapest-corridor cost over the first three seconds of a retrieval.

## 3.3 Episode-Level Feature Construction

Frame-level topology signals are aggregated to episode signatures: local pressure and nearest-support structure, corridor costs by lane, first-3-second closure, compactness/funnel tendencies, role-aware support geometry (F1/F2/F3 distances and pressure shares), D-pinch support count, and score-state context.

## 3.4 Archetypes and Validation Design

Archetypes are obtained with k-means in pressure-role feature space ($k = 4$). The supervised prediction target is **turnover in defensive zone**. Denial (turnover + forced dump-out) is used as the process lens for mechanism interpretation and coaching deployment.

Validation compares pressure-only and pressure+role models against nearest-distance and defender-count baselines using stratified cross-validated AUC.

# 4 Results

## 4.1 Retrieval Geography and Pressure Sanity Check

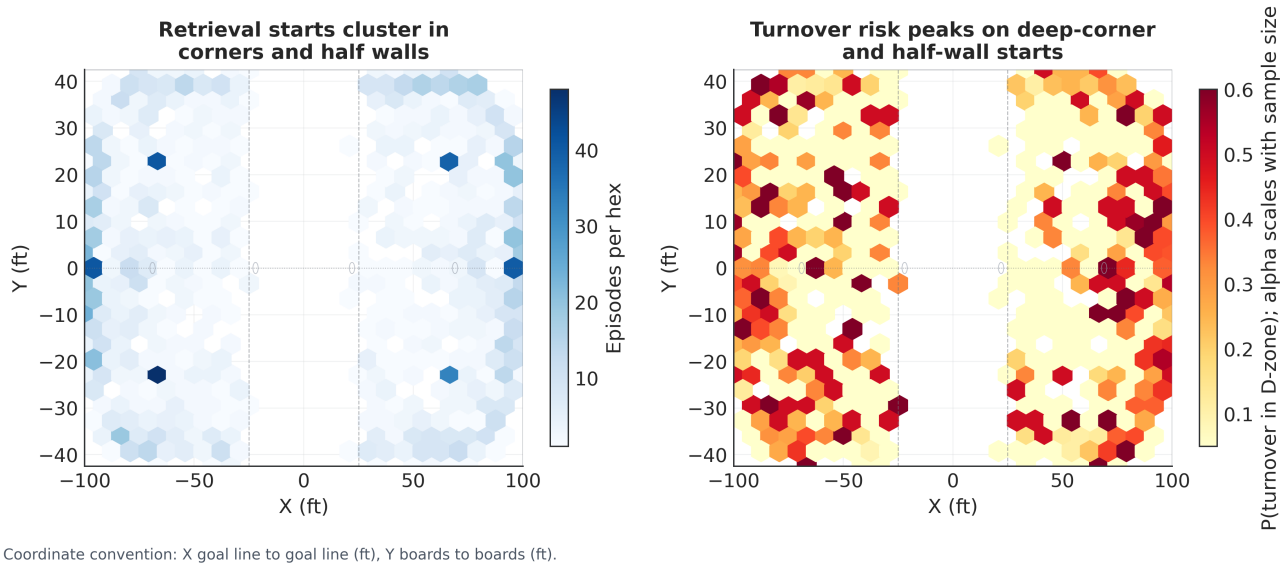


Figure 1: Episode start density (left) and turnover probability by start location (right, alpha scaled by sample size).

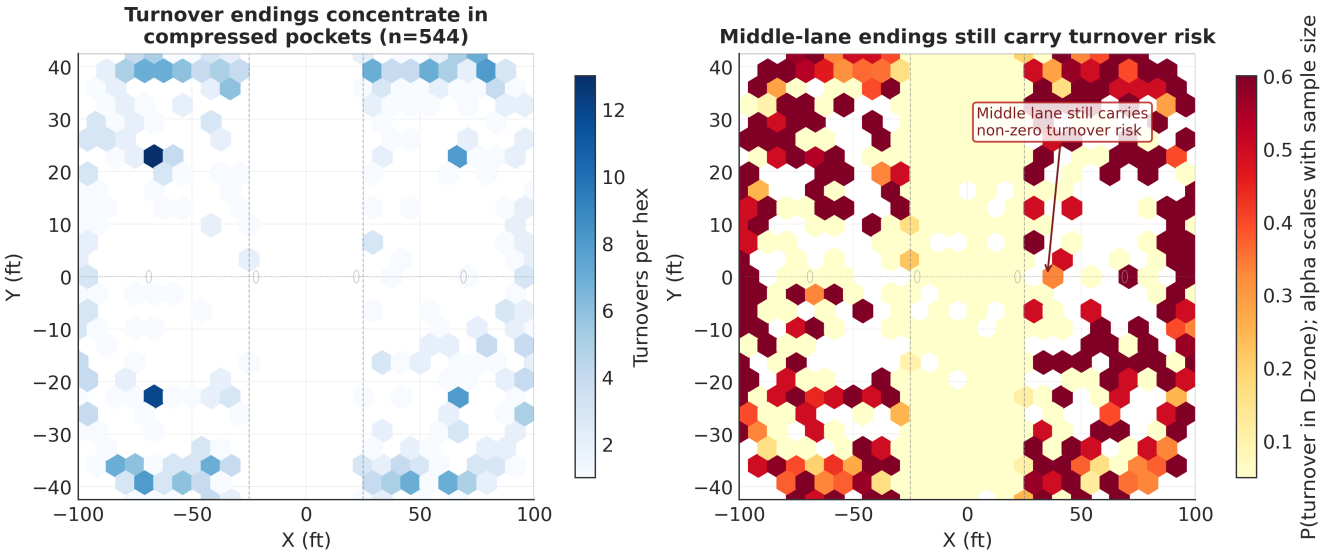


Figure 2: Turnover end-density (left) and turnover probability by episode end location (right, alpha scaled by sample size).

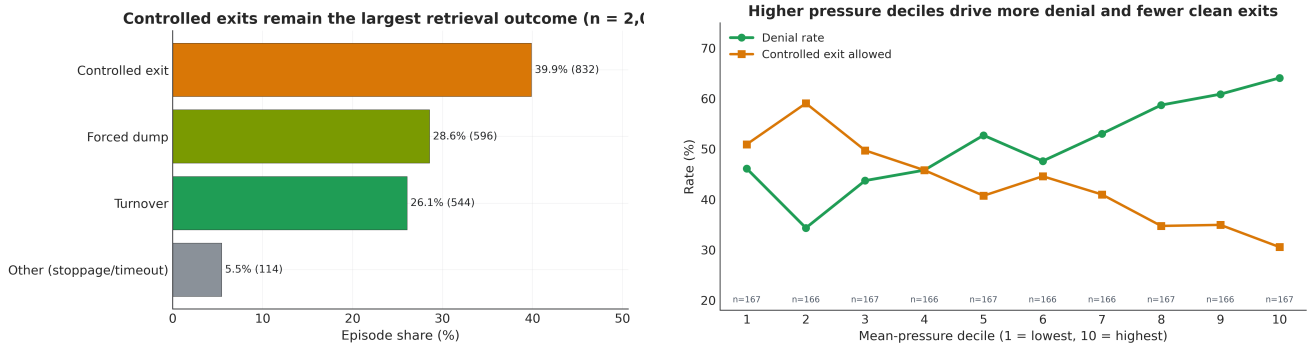


Figure 3: Outcome mix (left) and pressure-response sanity check (right). Denial rises from 46.1% to 64.1% across pressure deciles 1 to 10.

Pressure-response is strongest at the high end: top pressure-quartile denial is 62.6% versus 46.7% outside that quartile, while controlled exits fall from 46.8% to 32.4%. Geography figures also show that middle-lane areas are not turnover-free, despite concentration in deep-corner retrieval space.

## 4.2 Mechanistic Illustration (Single Episode)

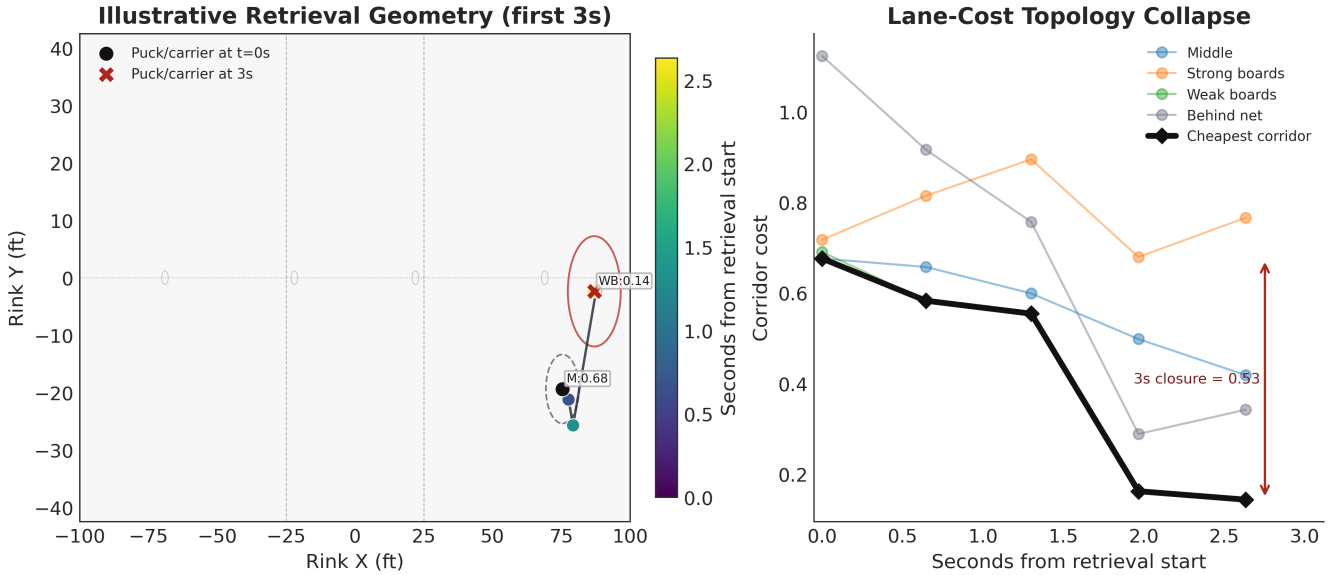


Figure 4: Illustrative retrieval sequence over first 3 seconds. The cheapest corridor is explicitly tracked and collapses by 0.53 in this turnover rep.

This panel provides direct intuition for the topology formulation: denial is not "high pressure" alone, but the time-ordered collapse of the cheapest lane while pressure layers remain connected.

## 4.3 Archetypes and Relative Topology

Table 2: Coach-Native Archetype Outcome Signatures

| Archetype | n | Turnover % | Forced Dump % | Controlled Exit % |
|---|---|---|---|---|
| Tight Squeeze (C0) | 157 | 33.1 | 29.3 | 33.1 |
| Light Pressure (C1) | 312 | 16.7 | 24.7 | 52.9 |
| Spread Support (C2) | 624 | 26.1 | 20.7 | 45.8 |
| Heavy Disrupt (C3) | 476 | 28.8 | 27.1 | 38.2 |

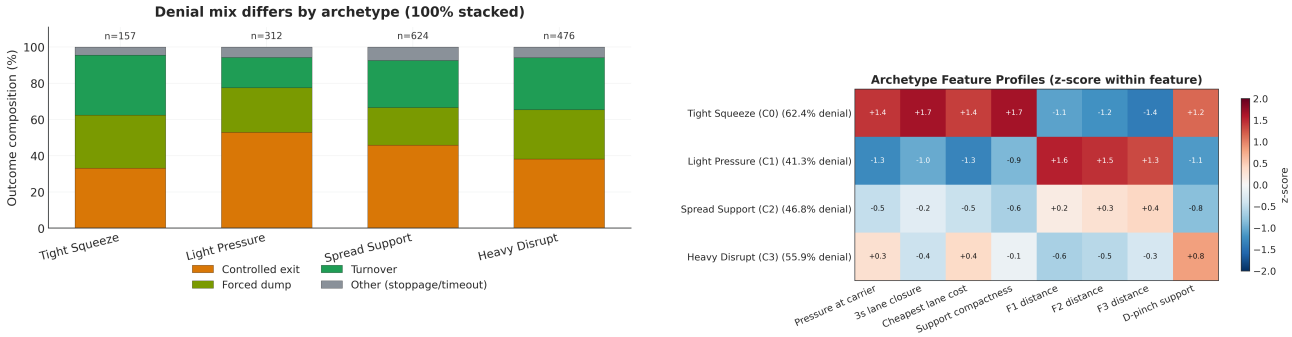


Figure 5: Archetype outcome composition (left, 100% stacked) and archetype feature profiles (right, z-score within feature).

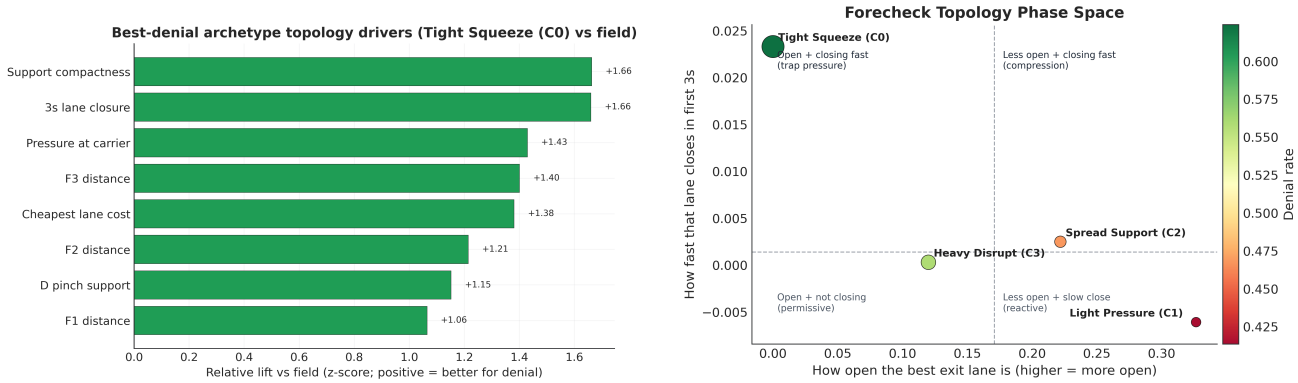


Figure 6: Relative lift profile for the best denial archetype (left) and topology phase-space interpretation (right).

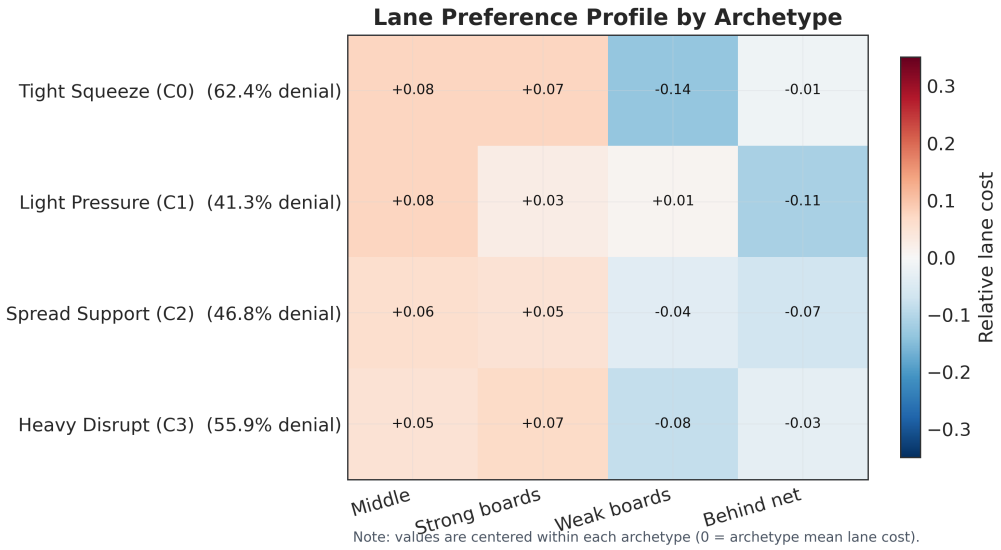


Figure 7: Lane preference profiles by archetype (values centered within archetype; 0 = archetype mean lane cost).

Tight Squeeze and Heavy Disrupt form the strongest denial structures. Light Pressure is the most permissive archetype and allows the highest clean exits. Across panels, high-denial archetypes pair pressure with support compactness and lane closure rather than pressure alone.

## 4.4 Role-Layering and Score-State Context

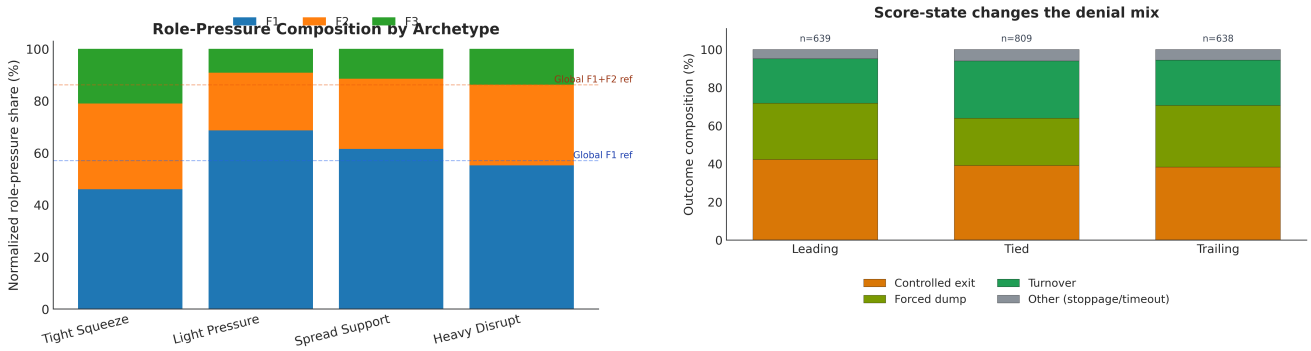


Figure 8: Role-pressure composition by archetype (left) and score-state denial composition (right).

Global normalized role composition is approximately F1 59.6%, F2 27.8%, and F3 12.6%. High-denial structures generally maintain stronger F2/F3 contribution integrity than low-pressure structures.

## 4.5 Closure Dynamics and Predictive Performance

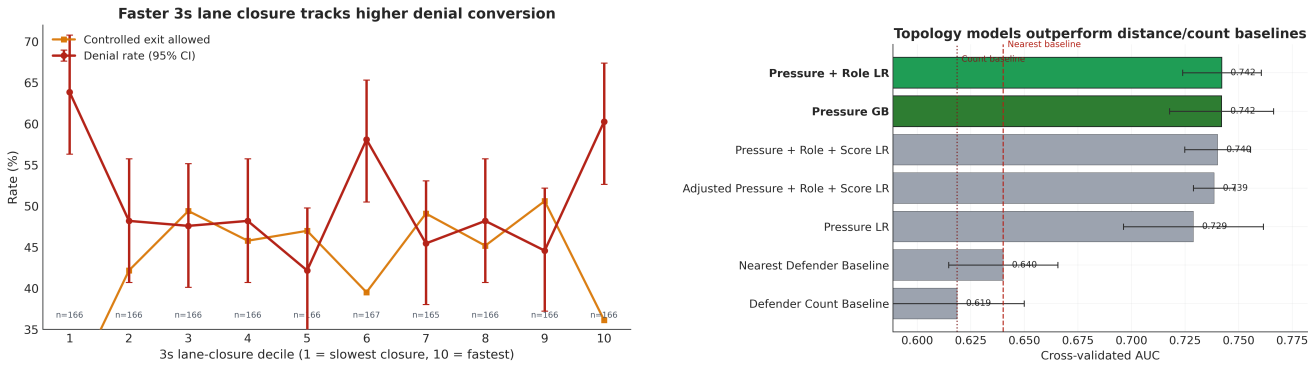


Figure 9: Closure-response relationship with 95% CI by decile (left) and model performance comparison (right).

Table 3: Turnover Prediction Performance (Primary Supervised Endpoint)

| Model | AUC Mean | AUC Std | n Obs |
|---|---|---|---|
| Pressure + role logistic | 0.742 | 0.018 | 1,569 |
| Pressure gradient boosting | 0.742 | 0.024 | 1,660 |
| Pressure + role + score logistic | 0.740 | 0.015 | 1,569 |
| Adjusted pressure + role + score logistic | 0.739 | 0.010 | 1,569 |
| Pressure-only logistic | 0.729 | 0.033 | 1,660 |
| Nearest-defender baseline | 0.640 | 0.025 | 1,665 |
| Defender-count baseline | 0.619 | 0.031 | 1,665 |

Best-model lift over nearest-defender baseline is approximately +0.102 AUC, indicating meaningful incremental signal from topology and role-aware structure.

## 4.6 Team Process Benchmarking

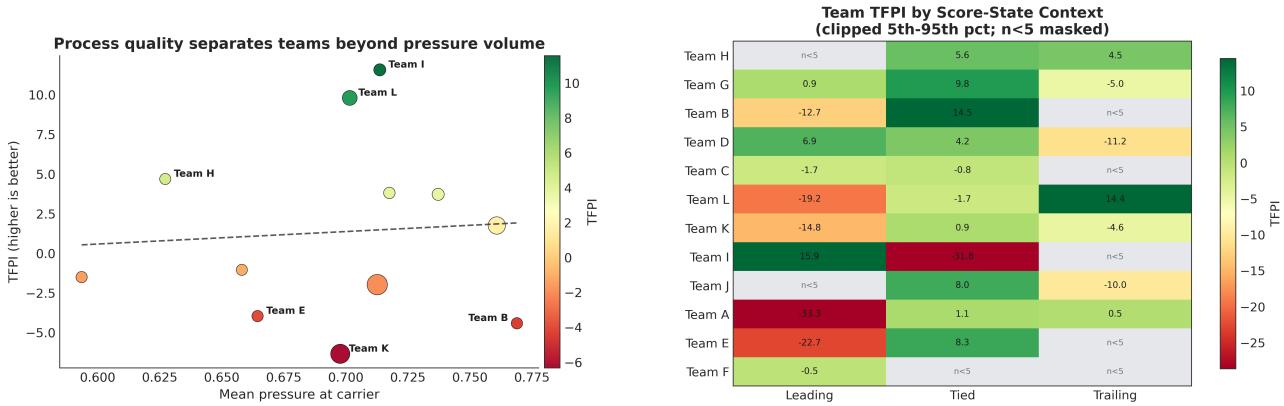


Figure 10: Team Forecheck Pressure Index (TFPI) overall (left) and by score-state context (right, clipped 5th-95th percentile and masked for sparse cells).

Team-level TFPI is defined as

$$\text{TFPI} = 100 \cdot (\text{TurnoverRate} + 0.5 \cdot \text{ForcedDumpRate} - \text{ControlledExitRate}).$$

In this sample, Team I (11.6) and Team L (9.8) are highest; Team K (-6.3), Team B (-4.4), and Team E (-4.0) are lowest. The 0.5 weight treats forced dump-outs as denials with lower direct transition value than immediate turnovers.

# 5 Discussion and Operational Deployment

The central empirical result is that denial is best explained by *lane-removal geometry*, not nearest-defender distance alone. In practice, this supports a deployment pattern where staffs pre-scout retrieval sectors with archetype tendencies, then tag clip review around first-3-second lane closure and F2/F3 layer continuity.

Because the framework is process-oriented and auditable, it is directly usable for line matching, role accountability, and within-team benchmarking under score-state context.

# 6 Limitations

The framework remains subject to four constraints: inferred role labels require manual video audit; acceleration and heading estimates inherit tracking noise; segmentation is rule-based rather than learned from tagged possession chains; and supervised validation targets event occurrence rather than downstream transition chance value.

# 7 Conclusion

Forechecking success on retrievals is better represented as a continuous topology of pressure, corridor availability, and closure dynamics. Retrieval Denial Topology produces interpretable mechanism-level signals and substantial predictive lift over standard distance/count baselines. The current system is suitable for practical coaching workflow integration and can be extended to transition-value modeling in future work.

# A Supplementary Cluster Sensitivity

## A.1 DBSCAN Sensitivity (Supplementary)

Table 4 summarizes the best DBSCAN settings by realized cluster count (all runs use min_samples = 5). Higher realized cluster counts are associated with large noise assignment and highly concentrated non-noise mass.

Table 4: DBSCAN Best Setting by Realized Cluster Count

| Realized Clusters | Eps | Noise % | Largest Cluster % | Silhouette (non-noise) | n Obs |
|---|---|---|---|---|---|
| 7 | 1.4 | 93.1 | 64.8 | 0.061 | 1,569 |
| 6 | 1.5 | 85.9 | 90.1 | -0.141 | 1,569 |
| 5 | 2.2 | 26.6 | 98.7 | -0.010 | 1,569 |
| 4 | 1.9 | 46.2 | 97.7 | 0.211 | 1,569 |
| 3 | 2.4 | 18.7 | 99.3 | 0.386 | 1,569 |
| 2 | 2.6 | 13.4 | 99.6 | 0.406 | 1,569 |

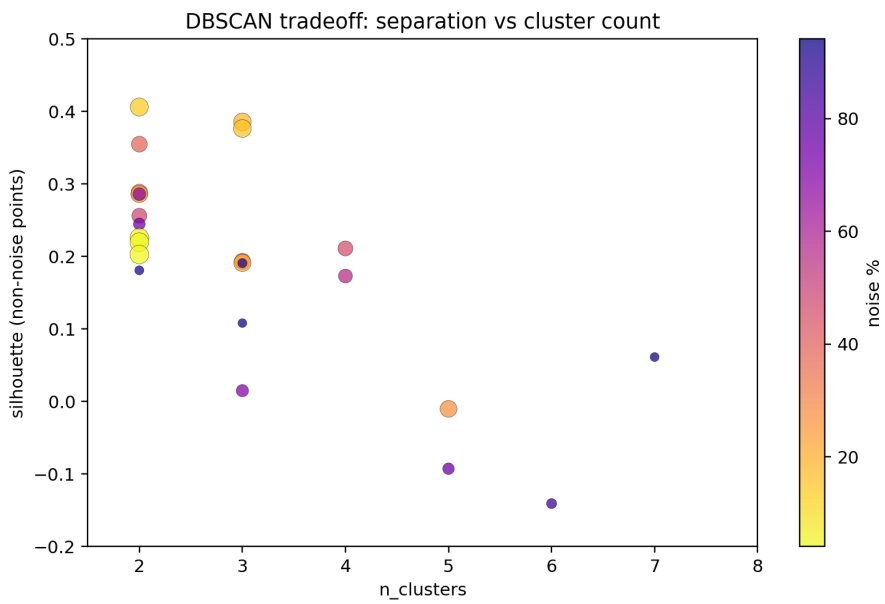


Figure 11: DBSCAN tradeoff surface across noise share and silhouette.

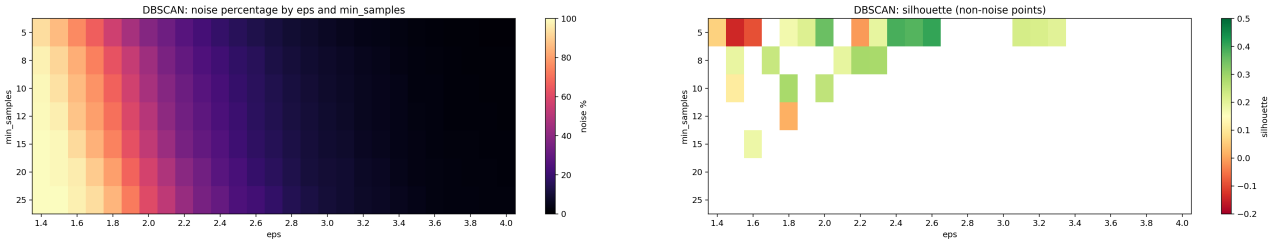


Figure 12: DBSCAN sensitivity heatmaps for noise assignment (left) and non-noise silhouette (right).

## A.2 GMM Sensitivity (Supplementary)

Table 5 summarizes best-component selections across covariance structures under BIC optimization.

Table 5: Best GMM Configuration by Covariance Type

| Covariance | Components | BIC | Silhouette | CH Index | DB Index | Largest Weight |
|---|---|---|---|---|---|---|
| full | 6 | 29,825.0 | 0.023 | 215.3 | 3.095 | 0.303 |
| tied | 12 | 43,731.9 | 0.035 | 105.6 | 1.981 | 0.629 |
| diag | 12 | 55,536.4 | 0.005 | 152.0 | 2.534 | 0.156 |
| spherical | 12 | 68,905.4 | 0.068 | 196.2 | 2.575 | 0.134 |

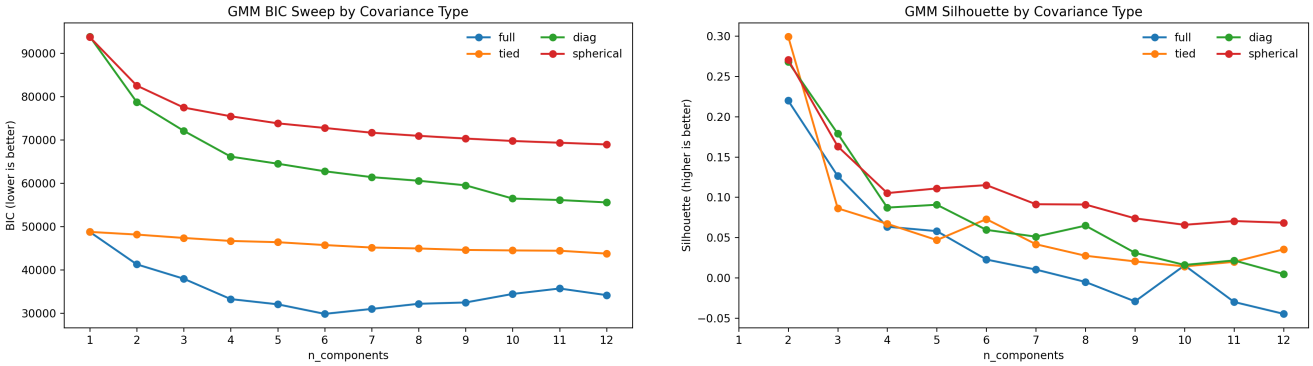


Figure 13: GMM model-order curves by covariance type: BIC (left) and silhouette (right).

# B Reproducibility and Artifact Paths

From repository root:

- `python projects/forechecking_pressure_topology/run_pipeline.py`
- `python projects/forechecking_pressure_topology/report_v1/scripts/make_figures.py`
- `powershell -ExecutionPolicy Bypass -File projects/forechecking_pressure_topology/report_v1/build_pdf.ps1`

Core project path: `projects/forechecking_pressure_topology/`